Arshiya, Puneetha, Neha, Pari, Parsa, Mohga

Fully dressed use cases:

<u>Arshiya Sabaa:</u>

Use Case UC1: Reserve Booking with Pre-Ordering

Primary Actor: Customer Stakeholders and Interests:

- Customer: Wants to secure a table and potentially save time by ordering food in advance.
- Restaurant: Wants to manage floor capacity and kitchen preparation timing.

Preconditions: Customer is logged into the "TablePlease" app.
Success Guarantee: Reservation is confirmed, and the pre-order is saved in the system.
Main Success Scenario:

1. Customer selects a restaurant and chooses "Reserve Booking".
2. Customer selects date, time, and party size.
3. System presents the option to "Order Beforehand" or "Order Once Seated".
4. Customer chooses "Order Beforehand" and selects items from the menu.
5. System validates table availability and saves the reservation and pre-order.
6. System provides a confirmation notification to the Customer.

Extensions:

- 3a. Customer chooses "Order Once Seated":
    1. System skips the menu selection and proceeds to confirm the reservation.

Special Requirements:

- The booking confirmation must be completed under 5 seconds.

Technology and Data Variation List:

- Menu items may include digital photos and allergy warnings.

Use Case UC2: Digital Bill Tracking and Payment

Primary Actor: Customer Stakeholders and Interests:

- Customer: Wants to see a "live" total to manage spending.
- Waiter: Wants to ensure the bill is accurate before final payment.

Preconditions: Customer is checked in at the restaurant.
Success Guarantee: Payment is processed, and the bill is closed.
Main Success Scenario.

1. Customer selects the "Live Bill" section in the app.
2. System displays all current charges, including pre-ordered items.
3. Customer orders additional items through the waiter or app.
4. System updates the live bill in real-time.
5. Customer selects "Pay Now" and uses a digital payment method.
6. System validates payment and sends a digital receipt.

Extensions:

- 5a. Payment Declined:
    1. System notifies the Customer and prompts for a different payment method.

Special Requirements:

- Payment transitions must be completed under 5 seconds.

Technology and Data Variation List:

- Supports Apple Pay, Google Pay, and manual credit card entry.

Use Case UC3: Manage Order for Pickup or Delivery

Primary Actor: Customer Stakeholders and Interests:

- Customer: Wants a simple way to order food for catering or personal pickup.
- Restaurant Staff: Wants to receive accurate order details and payment to begin preparation.
- Catering Manager: Needs advanced notice for large-scale catering orders.

Preconditions:

- Customer is logged into the TablePlease app.
- The restaurant has uploaded a pickup/delivery menu.

Success Guarantee: The order is successfully placed, payment is authorized, and the restaurant is notified.

Main Success Scenario:

1. Customer selects a restaurant and chooses the "Pickup/Delivery" option.
2. Customer browses the menu and adds items to their digital cart.
3. Customer selects the order type: "Order" or "Catering".
4. Customer selects a desired pickup or delivery time.
5. System displays the total bill, including any applicable fees.
6. Customer confirms the order and completes the digital payment.
7. System sends the order details to the Restaurant Dashboard.

Extensions:

- 3a. Catering Order selected:
    1. System prompts for additional details (e.g., number of guests, special setup instructions)
- 6a. Payment failure:
    1. System notifies the customer and requests an alternative payment method.

Special Requirements:

- The menu and checkout pages must load in under 5 seconds.
- The system must handle high-volume ordering during peak hours without performance degradation.

Technology and Data Variation List:

- Menu items are retrieved from the restaurant's profile database.
- Order can be placed via the mobile app or a web browser.

Use Case UC4: Track Order Stages

Primary Actor: Customer Stakeholders and Interests:

- Customer: Wants real-time updates on their order status to coordinate pickup or receive delivery.
- Kitchen Staff: Needs to update the system as they progress through the order.
- Delivery Driver: Needs to signal when the order is "In Transit" or "Delivered".

Preconditions:

- An order has been successfully placed and confirmed (see UC8).

Success Guarantee: The Customer is kept informed of the order status until fulfillment.

Main Success Scenario:
1. Customer navigates to the "Order Tracking" page.
2. System displays the current stage of the order: "Order Received".
3. Restaurant staff updates the status to "Preparing" on their dashboard.
4. System updates the Customer's tracking page in real-time.
5. Restaurant staff updates the status to "Ready for Pickup" or "Out for Delivery".
6. System sends a push notification to the Customer.
7. Once the order is received, the system marks the order as "Completed".

Extensions:
- 5a. Delay in preparation:
    1. Restaurant staff updates the estimated time.
    2. **System notifies the Customer of the updated arrival/pickup time.**

Special Requirements:
- The tracking page must refresh every 10 seconds to provide live updates.
- Status transitions (e.g., from "Preparing" to "Ready") must reflect on the Customer's device within 3 seconds.

Technology and Data Variation List:
- Status updates are triggered by staff via the Restaurant Dashboard.
- Delivery tracking may include a map view if the driver's GPS is enabled.

Parsa Mirlohi

Use Case UC5: Register Restaurant Account

Primary Actor

Restaurant Owner

Stakeholders and Interests

- **Restaurant Owner:** Wants to register the restaurant to access platform tools.
- **Platform Administrator:** Wants only verified businesses available in the system.
- **Customers:** Want accurate restaurant information.

Preconditions

- Restaurant owner has access to the TablePlease application.

Success Guarantee (Postconditions)

- Restaurant account is created.
- Verification request is recorded by the system.

Main Success Scenario

1. Restaurant owner selects “Register Restaurant.”
2. System presents restaurant registration form.
3. Restaurant owner enters required restaurant information.
4. Restaurant owner submits verification documents.
5. System validates entered information.
6. System creates a pending restaurant account.
7. System records verification request.
8. System confirms successful submission to the owner.

Extensions

5a. Missing or invalid information

- System displays validation errors.
- Restaurant owner corrects information.
- System resumes at Step 5.

6a. Duplicate restaurant detected

- System notifies owner that restaurant already exists.
- Registration process is canceled.

Special Requirements

- Registration submission shall complete within **5 seconds** under normal network conditions.

Technology and Data Variation List

- Verification documents may include business license or tax identification files.
- Information may be entered with mobile or web interface.

Frequency of Occurrence

- Occurs once per restaurant registration.

Use Case UC6: Manage Restaurant Profile

Primary Actor

Restaurant Admin

Stakeholders and Interests

- **Restaurant Admin:** Wants accurate restaurant information displayed to customers.
- **Customers:** Want reliable restaurant details before booking.

- **Platform:** Requires updated and consistent data.

Preconditions

- The restaurant account has been verified.
- Restaurant admin is authenticated in the system.

Success Guarantee (Postconditions)

- Updated restaurant profile information is saved.
- Updated information becomes visible to customers.

Main Success Scenario

1. Restaurant admin logs into the restaurant dashboard.
2. System displays restaurant management options.
3. Restaurant admin selects “Edit Profile.”
4. System displays current restaurant information.
5. Restaurant admin modifies profile details.
6. Restaurant admin submits changes.
7. System validates updated information.
8. System saves updated profile.
9. System confirms successful update.

Extensions

7a. Invalid information entered

- System displays validation error message.
- Restaurant admin corrects information.
- System resumes at Step 7.

Special Requirements

- Profile updates shall propagate to users within **3 seconds**.

Technology and Data Variation List

- Profile images may be uploaded from mobile or desktop devices.
- Text information may support multiple languages.

Frequency of Occurrence

- Occurs occasionally when restaurant information changes.

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Use Case UC7: Configure Table Layout and Seating

Primary Actor

Restaurant Admin

Stakeholders and Interests

- **Restaurant Admin:** Wants accurate seating configuration.
- **Host Staff:** Needs layout information for seating customers efficiently.
- **Customers:** Benefit from accurate wait-time estimates.

Preconditions

- Restaurant admin is logged into the dashboard.
- Restaurant profile exists.

Success Guarantee (Postconditions)

- Updated table layout is saved and available to staff.

Main Success Scenario

1. Restaurant admin opens the "Table Layout Editor."
2. System displays current table layout.
3. Restaurant admin adds or selects a table.
4. System prompts for table attributes.
5. Restaurant admin specifies seating capacity and table type.
6. Restaurant admin saves layout configuration.
7. System validates layout information.
8. System updates restaurant seating configuration.
9. System confirms successful update.

Extensions

7a. Invalid layout configuration detected

- System highlights conflicting table placement.
- Restaurant admin adjusts layout.
- System resumes at Step 7.

Special Requirements

- Layout updates shall appear on all staff dashboards within **3 seconds**.

Technology and Data Variation List

- Tables may include attributes such as booth, outdoor seating, or accessible seating.
- Layout editing may use drag-and-drop or manual input.

Frequency of Occurrence

- Occurs infrequently, typically during setup or redesign.

Use Case UC8: Manage Waitlist and Table Status Dashboard

Primary Actor

Host Staff

Stakeholders and Interests

- **Host Staff:** Wants real-time visibility of tables and waitlist.
- **Restaurant:** Wants efficient seating turnover.
- **Customers:** Expect accurate wait-time updates.

Preconditions

- Staff member is authenticated.
- Restaurant layout has been configured.

Success Guarantee (Postconditions)

- Table status and waitlist information are updated system-wide.

Main Success Scenario

1. Host staff logs into the dashboard.
2. System displays waitlist and table status overview.
3. Host selects a table.
4. System displays available status options.
5. Host updates table status.
6. System records updated table status.
7. System recalculates waitlist estimates.
8. System refreshes dashboard for all active staff users.

Extensions

5a. Invalid status transition

- System rejects the update.
- System displays error message.
- Host selects a valid status.

8a. Network synchronization delay

- System retries update synchronization.
- System notifies staff if delay persists.

Special Requirements

- Dashboard updates shall synchronize across devices within **3 seconds**.

Technology and Data Variation List

- Dashboard may be accessed via tablet, desktop, or mobile device.
- Status updates may be triggered manually by staff.

Frequency of Occurrence

- Occurs continuously during restaurant operating hours.

Puneetha Mallarapu:

Use Case UC9: Online Ordering for Event Reservations.

Primary Actor: Customer Stakeholders and Interests:

- Customer: Wants to reserve a larger spot for an event and place an order ahead of time for convenience.
- Restaurant Owners: Wants to prepare ahead of time for larger events and ensure customer satisfaction.

Preconditions:

- Customers would need to have access to the app and a designated log in.
- Event details and availability are stored in the system with a higher priority.

Success Guarantee: There exists a page on the app where large-scale event reservations and online ordering can be made and successfully submitted to the restaurant.

Main Success Scenario:

1. The customer opens the app
2. The customer navigates to the reservation page.
3. TThe customer selects their event.
4. The system displays details, time slots, and ordering options.
5. The customer selects reservation time and menu item for ordering ahead.
6. Customer confirms the reservation and order.

7. The system processes the reservation and payment.
8. The system sends confirmation to the customer.

Extensions:
- 4a. In case of selected time slots no longer being available, the system displays a message to the customer and prompts a new selection.
- 8a. In case the payment fails, the system displays a message to the customer and asks them to retry with another payment method.

Special Requirements:
- The event reservation page must load in under 5 seconds.
- Have the order ahead feature update every 5 minutes in case of changes to the menu items.

Technology and Data Variation List:
- Event availability data is stored and retrieved from a local database.

Use Case UC10: Page for all Available Restaurants/Cuisines

Primary Actor: Customer Stakeholders and Interests:
- Customer: Wants to be able to browse all available restaurants and cuisine options to choose where they want to eat.
- Restaurant Owners: Wants to be properly represented and visible for potential future customers.

Preconditions:
- Customers would need to have access to the app to browse an account with all necessary personalizations.
- Restaurants would have listed and put all their information into the database.

Success Guarantee: There exists a page on the app with all restaurant names, cuisines, and other valuable information for the customer to look and browse through.

Main Success Scenario:

9. The customer opens the app
10. The customer navigates to the browsing page
11. The system displays all the restaurant names with their specific cuisines.

Extensions:
- 3a. In case of closings or holidays, the system displays a message indicating that there are no restaurants available.

Special Requirements:
- The restaurant page must load in under 5 seconds.

Technology and Data Variation List:
- The restaurant data is stored and retrieved from a local database.
- Cuisine categories may be updated or allow restaurants to choose multiple options.

Use Case UC11: Recommendation Service

Primary Actor: Customer Stakeholders and Interests:
- Customer: Wants personalized suggestions based on their preferences and previous excursions.

- Restaurant Owners: Wants to target customers who are more likely to enjoy their food or more likely to reserve a spot.

Preconditions:
- Customers would need to have access to the app and a designated log in.
- Customers would have reserved or rated restaurants (on the app) in the past that are saved.

Success Guarantee: There exists a page on the app with personalized restaurant recommendations for the customer.
Main Success Scenario:
1. The customer opens the app.
2. The customer navigates to the recommendations page.
3. The system analyzes the customer's past reservations and reviews.
4. The system generates recommendations.
5. The system displayed the recommendations to the customer.

Extensions:
- 3a. In case of no prior reservation data, the system displays popular or trending restaurants in the area.

Special Requirements:
- The recommendation page must load in under 5 seconds.

Technology and Data Variation List:
- The past reservation data is stored and retrieved from a local database.
- Recommendation services might be done using machine learning.

Use Case UC12: Rating your Experience

Primary Actor: Customer Stakeholders and Interests:
- Customer: Wants to provide feedback about their experience at certain restaurants.
- Restaurant Owners: Wants to garner feedback to improve their service or to gain more publicity and attract new customers.

Preconditions:
- Customers would need to have access to the app and a designated log in.
- Customers would have reserved or been to a restaurant (on the app) in the past.

Success Guarantee: There exists an option on the app for customers to rate and provide feedback after a reservation or restaurant visit that saves successfully.
Main Success Scenario:
1. The customer opens the app.
2. The customer navigates to their past visits or reservations.
3. The customer selects a specific visit/restaurant.
4. The system displays the rating page.
5. The customer submits a rating and commentary.
6. The system stores the rating and later displays it on the restaurant's rating page.

Extensions:
- 5a. In case of a customer canceling their rating, the system deletes the changes and returns to the previous page.

Special Requirements:
- The rating page must load in under 5 seconds.
- The rating should take at max 5 minutes to process before being displayed on the restaurant's rating page.

Technology and Data Variation List:
- The reservation data or past visit data is stored and retrieved from a local database.
- Ratings will be numeric from 1-5 stars and include comments from the customer.
- Feedback data is stored locally and is retrieved quickly for displaying.

<u>Pari Rana:</u>

Use Case UC13: Create Account & Login

Primary Actor: Customer
Stakeholders and Interests:
- Customer: wants fast access and secure account
- System: wants verified users and protected data

Preconditions:
Customer has downloaded the TablePlease App.

Success Guarantee:
Customer is successfully logged in and can access all the features.

Main Success Scenario:
1. Customer opens the TablePlease app
2. Customer selects "Sign up" or "Log in"
3. Customer enters phone number or email and password
4. System validates
5. System logs the Customer onto the dashboard

Extensions:
3a. Invalid credentials entered:
System displays an error message and prompts Customer to re-enter information.

3b. Customer selects "Forgot Password":
System sends password reset instructions by email or SMS

Special Requirements:
Account authentication must be completed under 5 seconds.

Technology and data variation list:
Login may be completed using phone number or email address
Two-factor authentication

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Use Case UC14: Manage Personalized Profile

Primary Actor: Customer
Stakeholders and Interests:

- Customer: Wants saved preferences(party size, seating, allergies, and accessibility needs)

Preconditions:
Customer is logged into the TablePlease app.

Success Guarantee:
Customer profile preferences are saved and applied to future reservations
Main Success Scenario:

1. Customer navigates to the “Profile” section.
2. System displays profile fields that can be edited
3. Customer updates such fields
4. Customer selects save changes
5. System validates and stores the information
6. System confirms successful update

Extensions:
4a. Missing or invalid information is entered
System highlights incorrect field and asks for correction
4b. Customer cancels update
System discards changes and returns to profile page

Special Requirements:
Profile updates must show in the system within 3 seconds

Technology and data variation list:
Preference data is stored in the user profile database.
Allergy and accessibility info may be selectable options or manual entries

Use Case UC15: Automated Reservation Reminders & Cancellation Notifications

Primary Actor: System
Stakeholders and Interests:

- Customer: Wants timely reminders to avoid missing reservations
- Restaurant: Wants to reduce no shows and improve scheduling

Preconditions:
Customer has an upcoming reservation

Success Guarantee:
Customer receives a reminder and/or cancellation notification successfully

Main Success Scenario:

1. System detects an upcoming reservation within the reminder time frame
2. System generates a reminder notification
3. System sends notification to Customer via push notification, email or SMS
4. Customer receives reminder

1. If reservation is cancelled:
2. System generates cancellation notification
3. System sends notification to Customer

Extensions:
3a. Notification delivery failure:
System retries sending notification

3b. Customer disables notification:
System does not send reminder

Special Requirements:
Notification must be delivered within the designated window such as 24 hours before the reservation

Technology and data variation list:
Notifications may be delivered via push notification, SMS, or email.

Use Case UC16: View Reservation History & Favorite Restaurants

Primary Actor: Customer
Stakeholders and Interests:

- Customer: Wants easy access to past and upcoming reservations for future booking
- Restaurant: Wants to encourage repeat visits

Preconditions:
Customer is logged into the TablePlease app.

Success Guarantee:
Customer can view reservation history and either favorite or unfavorite a restaurants.

Main Success Scenario:

1. Customer navigates to the “Reservations” page
2. System displays upcoming and past reservations
3. Customer selects a reservation to view details
4. Customer selects restaurant and chooses “Add to Favorites”
5. System updates favorites list
6. System confirms the action

Extensions:
2a. No reservations exist:
System displays “No reservations found”

4a. Restaurant already in favorites:
System removes restaurant from favorites list if selected again

Special Requirements:
Reservation history page must load within 5 seconds.

Technology and data variation list
Reservation history is retrieved from the system database.
Favorites list is stored in the user profile database.

Mohga Dawelbeit:

Use Case UC17: Checking in Via TablePlease App

Primary Actor: Customer
Stakeholders and interests:

- Customer: Wants to check in at the restaurant quickly
- Restaurant: Wants to get customers checked in quickly and effectively, so there isn't a holdup

Preconditions: Customer has made a reservation and successfully logged onto the "TablePlease" app.
Success Guarantee: Customer's reservation has been identified. Their table preferences have been taken into account. The customer was checked in. The customer was shown to their table.
Main Success Scenario:

1. Customer arrives at their restaurant.
2. Customer logs onto TablePlease app.
3. Customer navigates towards the "Check in" button.
4. Customer selects "I am here," and is taken to a screen that displays "You are all checked in! Please show your confirmation ID/ barcode to the host station."
5. Customer walks into restaurant and gives the barcode to those working at the host station.
6. A host scans the barcode.
7. The host shows the customer to their table.

Extensions:

- 6a. Bar code could not be read:
    1. System rejects the barcode
    2. Host responds to this error
        - An ID was provided by the app after the customer checked in
        - The Host manually enters the ID provided

Special Requirements:

- The "You are all checked in! Please show your confirmation ID/ barcode to the host station." message should be displayed less than 3 seconds after the customer checks in

Technology and Data Variation List:

- 6a. The reservation information can be processed through a laser scanner or manually by a keyboard

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Use Case UC18: Checking for Table Updates

Primary Actor: Customer
Stakeholders and interests:

- Customer: Wants to know when their table will be available

Preconditions: Customer has made a reservation and has successfully logged onto the "TablePlease" app.
Success Guarantee: Customer's reservation has been identified. Their table preferences have been taken into account. The queue of individuals demonstrates the number of individuals ahead of the customer. The customer's wait time to receive a table was updated.
Main Success Scenario:

1. Customer arrives at their restaurant.
2. Customer logs onto TablePlease app.
3. Customer navigates towards the "Check in" button.
4. Customer selects "I am here," and is taken to a screen that displays "You are all checked in! Please show your confirmation ID/ barcode to the host station."
5. Customer clicks on the "View Wait Time" Button.
6. The app displays the message "There is no wait time. Your table is ready."
7. The Customer gets their barcode scanned and is shown to their table.

Extensions:

- 6a. Wait time is 10min:
    1. App displays "Your wait time is 10 min. There are 2 individuals ahead of you."
    2. The customer exits the app.
    3. The customer goes back into the app, 10 min later, and checks if their table is ready
        - The app displays the message "There is no wait time. Your table is ready."

Special Requirements:

- The wait time screen should be updated every 10 seconds, so that the customer receives live updates regarding their table availability

Technology and Data Variation List:

- 6a. Hosts may be able to message users on the app, giving them more specific updates/ information

Use Case UC19: Viewing and Requesting Specific Table Arrangements

Primary Actor: Customer
Stakeholders and interests:

- Customer: Wants to be able to specify a table for their reservation

Preconditions: Customer has successfully logged onto the "TablePlease" app and has begun the process of making a reservation.
Success Guarantee: Information regarding available tables has been collected and displayed to the customer. Once the customer specifies their desired table, the number of available tables is updated.
Main Success Scenario:

1. Customer logs onto TablePlease app.

2. Customer navigates towards the Reservation button and creates a reservation.
3. Customer selects the “Table Specifications” button
4. Customer specifies that they would like to view the details for the tables available for today
5. The app displays the available tables and details regarding those tables, like where they are located in the restaurant, how many people can be seated at the table, etc.
6. Customer clicks on the specific table they would like for their reservation
7. Customer clicks the “Update Preferences" button
8. Customer receives table that they requested

Extensions:

- 2a. : Customer does not make a reservation
    1. Customer selects the “Table Specifications” button
    2. Customer specifies that they would like to view the details for the tables available for today
    3. App displays the available tables, and customer clicks on the specific table they would like
    4. Customer clicks the “Update Preferences" button
    5. The app displays the message “No reservations have been made. Preferences may not be updated.”

Special Requirements:

- The list of available tables, under the “Table Specifications" section, should be updated every minute
- It should take less than 3 seconds to be taken to the Table Specifications page once the button has been clicked

Technology and Data Variation List:

- 5a. A visual representation of the restaurant and table placements is displayed for the customer to better understand table locations

Use Case UC20: Booking the restaurant for a private event

Primary Actor: Customer

Stakeholders and interests:

- Customer: Wants to be able to book the entire restaurant for their own event

Preconditions: Customer has successfully logged onto the "TablePlease" app.

Success Guarantee: Information regarding which restaurant the customer would like to book and when they would like to book it for has been collected. Customer is able to pay for the booking. Customer receives a receipt for the booking.

Main Success Scenario:

1. Customer logs onto TablePlease app.
2. Customer clicks on the Search icon, to search for their desired restaurant.
3. Customer Selects a Restaurant.
4. Customer selects the “Book Venue” button.
5. Customer Specifies which day they would like to book the venue for.
6. Customer successfully adds their payment method.
7. Customer successfully pays for the venue.

8. TablePlease displays the message “Your venue has been successfully booked!”
9. Customer receives a digital receipt for their booking.

Extensions:

- 6a. The payment information provided is incorrect
    1. TablePlease gives the customer an error and displays the message “Please provide correct payment information.”
    2. Customer correctly inputs their payment information

Special Requirements:

- The app should support at least three different payment methods.

Technology and Data Variation List:

- 2a. AI may be utilized to provide the user with restaurant recommendations based on their previous bookings

Neha Rudrakshala:

Use Case UC21: Viewing Rewards Based on Spending

Primary Actor: Customer

Stakeholders and Interests:

- Customer: Wants to see rewards earned based on how much money they have spent at restaurants.
- Restaurant: Wants to encourage repeated visits and reward loyal customers.

Preconditions:

- Customer is logged into the TablePlease app.
- Customer has made at least one purchase at a participating restaurant.

Success Guarantee:

- Customer can view their current reward balance and available rewards.

Main Success Scenario:

1. Customer opens the TablePlease app.
2. Customer navigates to the “Rewards” section.
3. System retrieves the customer’s purchase history.
4. System calculates reward points based on spending.
5. System displays the customer’s available rewards and reward balance.
6. Customer reviews available rewards or benefits.

Extensions:

3a. Customer has no qualifying purchases:

- The system displays the message: “No rewards available yet. Continue dining to earn rewards.”

Special Requirements:

- The rewards page must load within 3 seconds.

Technology and data variation list:

- Reward points are calculated using restaurant-defined reward rules stored in the system database.

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Use Case UC22: Apply Loyalty Tier Benefits

Primary Actor: Customer

Stakeholders and Interests:

- Customer: Wants to receive additional discounts or benefits for being a loyal customer.
- Restaurant: Wants to incentivize repeat customers through loyalty tiers.

Preconditions:

- Customer is logged into the TablePlease app.
- Customer has accumulated spending or reward points.

Success Guarantee:

- Customer receives the correct loyalty tier and associated benefits.

Main Success Scenario:

1. Customer navigates to the "Rewards" section of the app.
2. System retrieves the customer's reward points and purchase history.
3. System evaluates the customer's loyalty tier level.
4. System assigns the appropriate tier to the customer.
5. System displays the tier benefits such as discounts or exclusive rewards.

Extensions:

3a. Customer does not qualify for a higher tier:

- System assigns the default loyalty tier and displays basic benefits.

Special Requirements:

- Tier updates must occur automatically after each qualifying purchase.

Technology and Data Variation List:

- Tier thresholds may vary depending on restaurant reward policies.

Use Case UC23: Restaurant Configure Reward Point System

Primary Actor: Restaurant Admin

Stakeholders and Interests:

- Restaurant Admin: Wants to define how customers earn reward points for purchases.
- Customer: Wants to earn points and redeem rewards when dining at the restaurant.

Preconditions:

- Restaurant admin is logged into the restaurant dashboard.
- Restaurant account has been verified.

Success Guarantee:

- Restaurant reward point system is saved and applied to customer purchases.

Main Success Scenario:

1. Restaurant admin logs into the restaurant dashboard.
2. Restaurant admin navigates to the "Reward Settings" section.
3. System displays reward configuration options.

4. Restaurant admin defines the reward rule (e.g., points per dollar spent).
5. Restaurant admin saves the reward configuration.
6. System validates and stores the reward system settings.
7. System applies the reward system to customer purchases.

Extensions:
6a. Invalid reward configuration entered:
- System displays a validation error and prompts the admin to correct the input.

Special Requirements:
- Reward system updates must be reflected in the customer app within 3 seconds.

Technology and Data Variation List:
- Reward systems may include point accumulation, percentage discounts, or promotional bonuses.

Use Case UC24: Search Restaurants by Name or Zip Code

Primary Actor: Customer
Stakeholders and Interests:
- Customer: Wants to quickly find restaurants near their location.
- Restaurant: Want their restaurant to be discoverable by potential customers.

Preconditions:
- Customers have access to the TablePlease app.

Success Guarantee:
- Customer receives a list of restaurants matching the search criteria.

Main Success Scenario:
1. Customer opens the TablePlease app.
2. Customer navigates to the search page.
3. Customer enters a restaurant name or zip code.
4. System searches the restaurant database for matching results.
5. System displays a list of restaurants that match the search query.
6. Customer selects a restaurant to view additional details.

Extensions:
4a. No restaurants match the search query:
- System displays the message “No restaurants found for this search.”

Special Requirements:
- Search results must load within 5 seconds under standard 4g/5g wi-fi conditions.

Technology and Data Variation List:
- Search results may be filtered by location, cuisine type, or restaurant rating.

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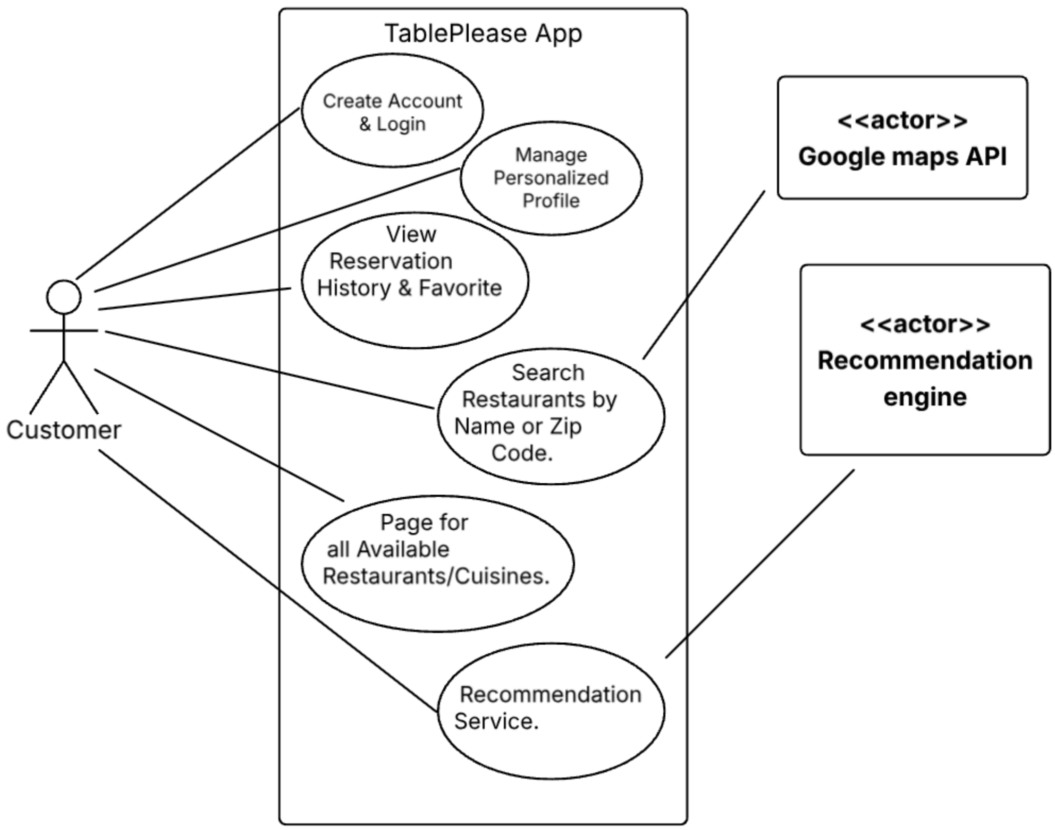
Use Case Diagrams:

1. Discovery & User Lifecycle Diagram- **Pari**



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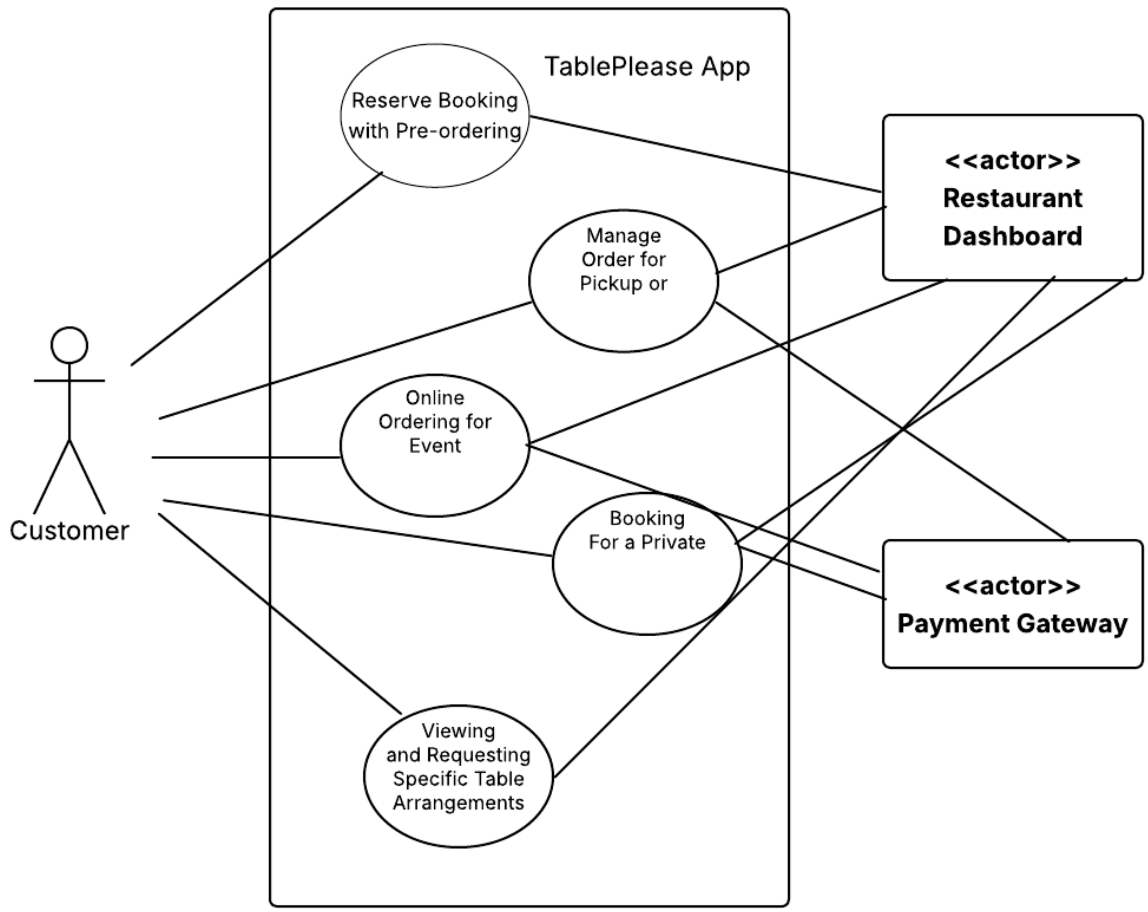
2. Reservations & Ordering Diagram - **Neha Rudrakshala**



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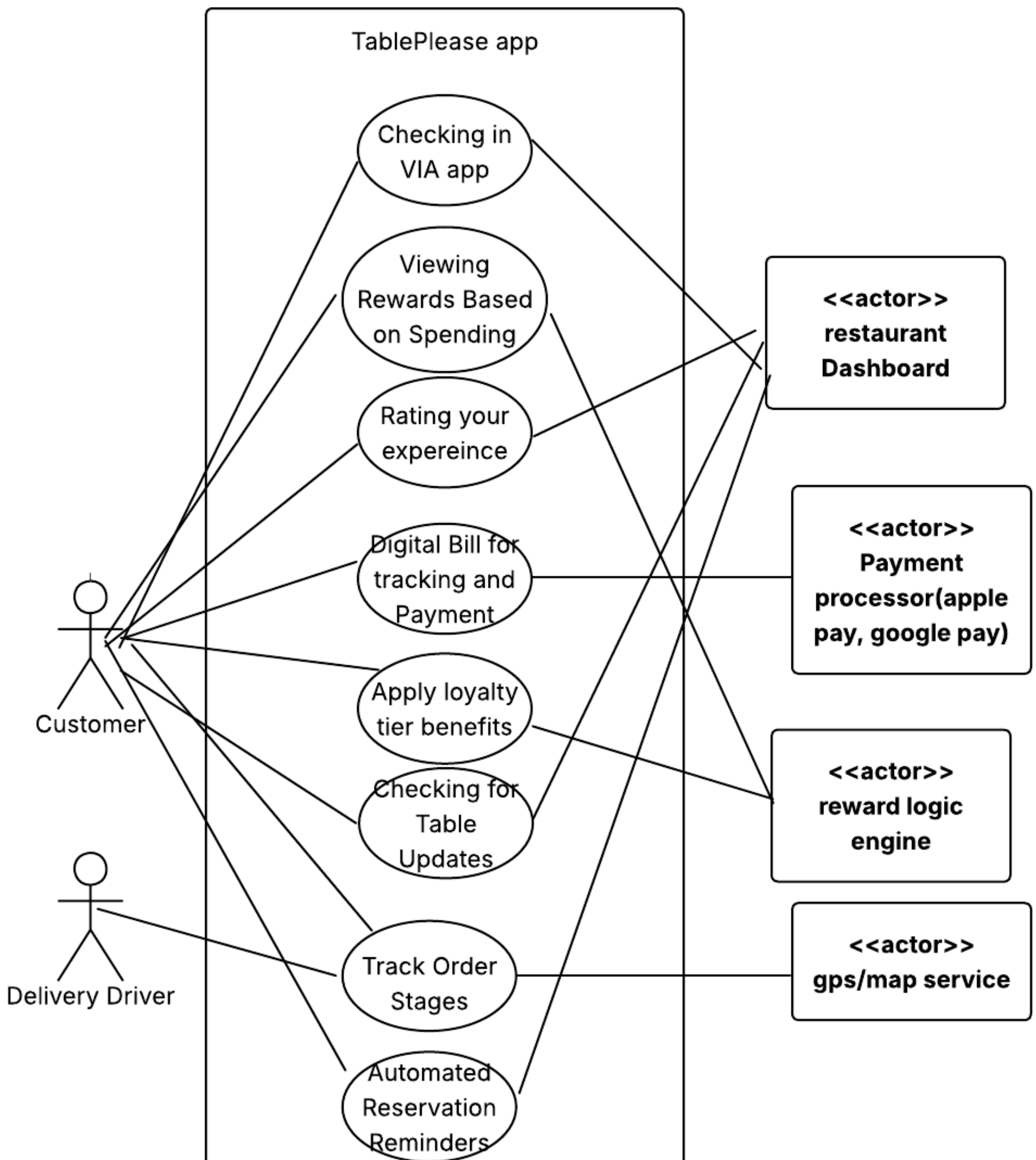
3. Dining Experience & Loyalty Diagram **- Arshiya Sabaa**



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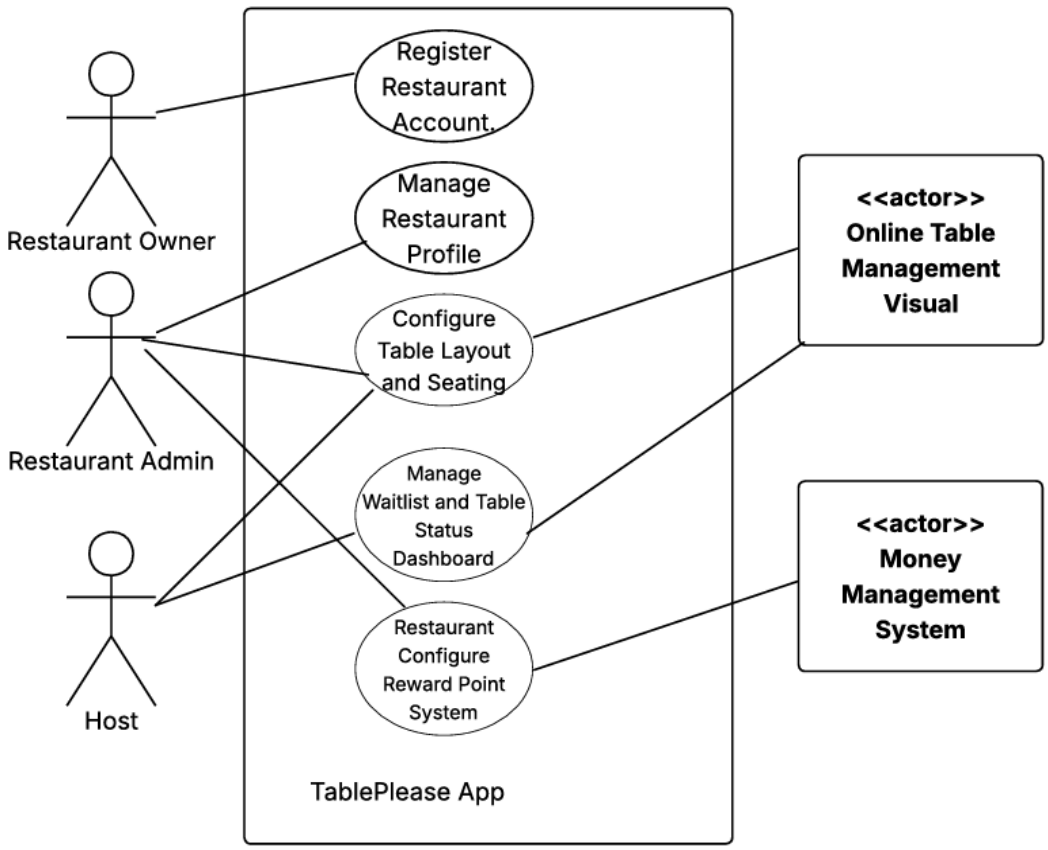
4. Restaurant Operations Diagram- **Mohga**



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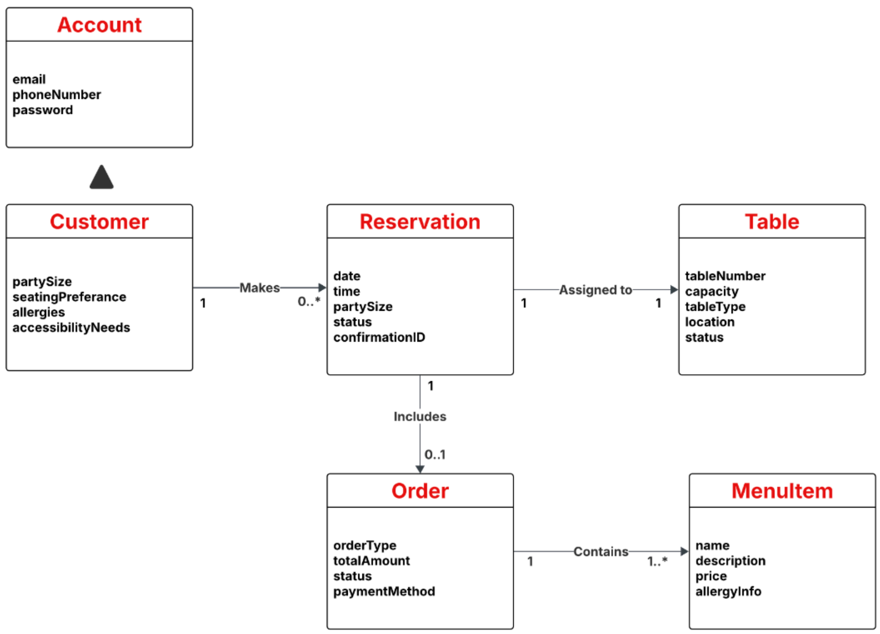
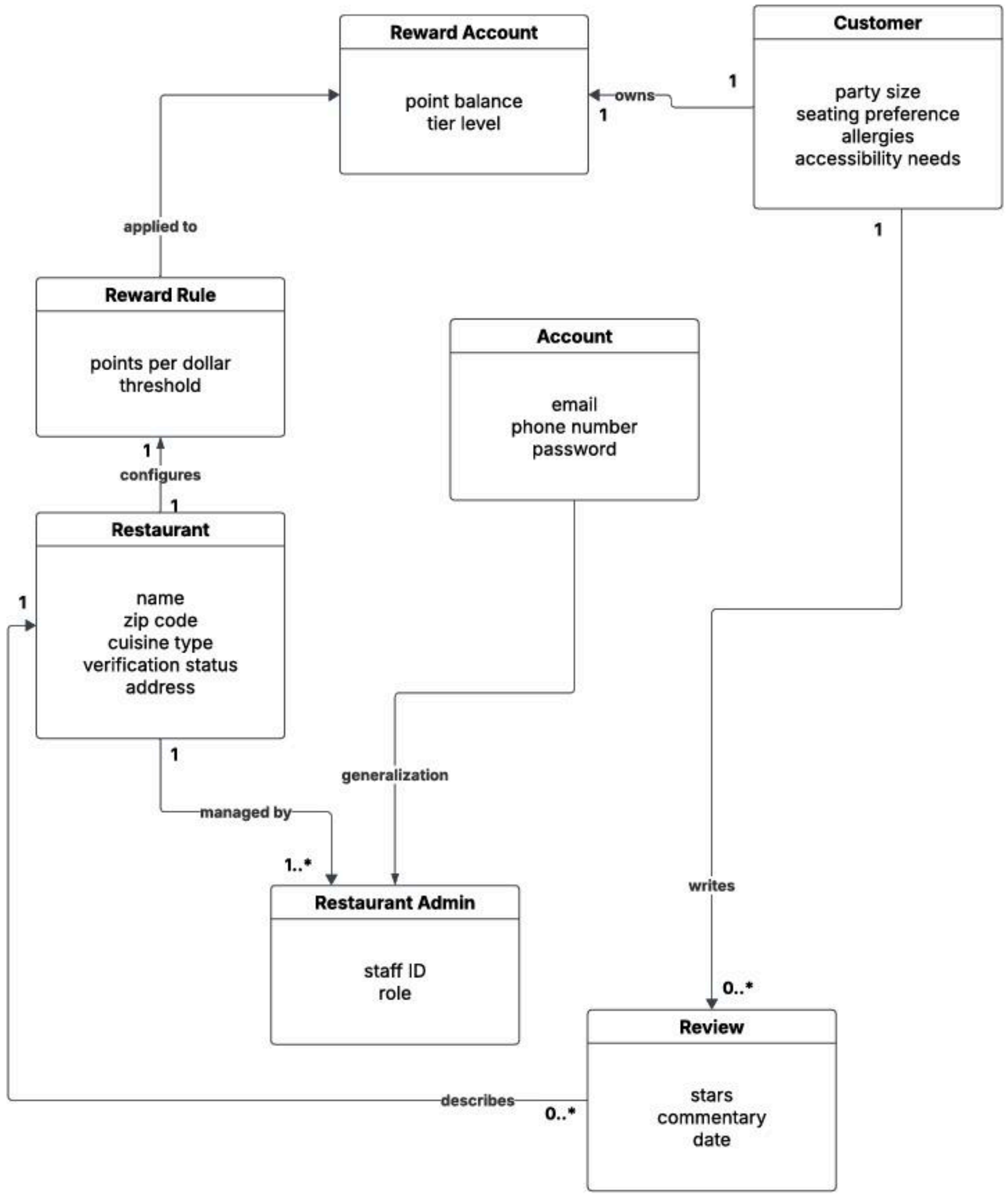
Conceptual Class Diagrams

1. Core Dining & Reservations - **Parsa Mirlohi**



2. Restaurant Management & Loyalty **- Puneetha Mallarapu**



Arshiya, Puneetha, Neha, Pari, Parsa, Mohga

Supplementary Specifications:

Parsa Mirlohi:

- The dashboard/system must maintain real time data consistency so that the table status, waitlist updates, and restaurant profile changes are synchronized across all active devices without conflicting information.

Arshiya Sabaa:

- The app must have low latency. Every functional requirement should load under 5 seconds, and transitions like confirming bookings and payments, must be completed under 5 seconds under standard 4g/5g wi-fi conditions.

Neha Rudrakshala:

- The system must provide notifications to customers about short-term rewards, promotions, or discounts created by restaurants. Notifications must be delivered through in-app alerts or push notifications and should appear within one minute after the restaurant activates the promotion.

Pari Rana:

- The system must ensure secure handling of user data, which include login credentials, reservation details, and payment information. All sensitive information must be encrypted to prevent unauthorized access.

Puneetha Mallarapu:

- Have the order ahead feature update every 5 minutes in case of long wait times or temporary closures. The rating system for restaurants must also take at most 5 minutes to process before being displayed on the restaurant's rating page.

Mohga Dawelbeit

- The TablePlease app must be able to support at least 200 restaurants and their corresponding information.

Github Link: https://github.com/pmallarapu/CS3704Project.git